

Redis Github:

- <https://github.com/redis/redis>

Initial Steps in Porting Redis:

1. Make a directory that is meant specifically for you.
 - a. Ex: mkdir aquiles
2. Go into that directory
 - a. cd aquiles
3. Use the command "zopen generate" to specify the tool
 - a. How it is formatted:
 - i. Project Name
 - ii. Description
 - iii. License
 - iv. Source URLs
 1. Grab the SSH link from the project
 - v. Build Dependencies
 1. In the READ.me, the build dependencies should be listed
 - vi. Runtime Dependencies
 1. In the READ.me, the runtime dependencies should also be listed
4. CD into your project name
 - a. Ex: cd redisport
5. Go into the buildenv file using a text editor
 - a. This could be done with an editor outside of the UNIX system such as VS Studio, but it could also be done with Vim or nano.
6. Ensure the information there aligns with the build information from the official redis github.
 - a. The buildenv file defines the environment in which the project will be built and tested. Here are some aspects of this file:
 - i. Dependencies: Lists the libraries and tools necessary to build the project. May have specified specific versions.
 1. How it may look:
 - a. `export ZOPEN_STABLE_DEPS="make openssl
which pkgconfig check_python"`

- ii. Environment variables: Sets the variables needed for the build process like paths to tools and libraries
 - 1. export
ZOPEN_STABLE_URL="https://github.com/redis/redis.git"
 - iii. Build tools: Specifies the tools and compilers used for building the project
 - 1. export ZOPEN_COMP="CLANG"
 - iv. Scripts: Automate the process of building
 - 1. # bump: redis-version /REDIS_VERSION="(.*)"/
https://github.com/redis/redis.git|semver:*
 - 2. # REDIS_VERSION="V.R.M" # Specify a stable release
 - v. Configurations Settings: Contains settings that configure the build process such as optimization flags and compiler options
 - 1. zopen_pre_patch()
 - 2. {
 - 3. export CFLAGS="\$CFLAGS \$CPPFLAGS
-mzos-target=zosv2r5 -D__XPLAT -D_OPEN_THREADS=2
-D__thread= -DMAP_ANON=0"
 - 4. }
7. Save changes then exit out of buildenv
 - a. Use ctrl+shift+o to save the changes made to the file
 - b. Use ctrl+shift+x for exiting the file
 8. Build the project using zopen build -vv
 - a. -vv will allow for more detailed information about how the project is building

Committing changes to Github:

1. Cd into the main repository
 - a. Ex: cd aquiles/redisport/redis
2. Use git diff to take the changes made and put in your patch
 - a. Ex: git diff > ../patches/pr.patch

3. Create a new patch that has all the new changes:
 - a. Ex: `git add buildenv patches/pr1.patch`
4. (Optional) You could remove a previous patch:
 - a. Ex: `rm patches/PR1.patch`
5. Specify the SSH key you are using to get ready to commit:
 - a. Ex: `export GIT_SSH_COMMAND="what you have as the ssh"`
6. Commit your changes using "git commit"
 - a. This will take you to a screen asking you to add a comment for that commit.
 - b. Once you are done press escape to get out of insert mode
 - c. From there press ":" by doing Shift+; which will put you into command mode
 - d. Type wq to save then quit.
 - i. This way the commit goes through
7. (Optional) Fix the SSH authorization error since same user is shared:
 - a. First use this: `export GIT_SSH_COMMAND="/bin what you have as the ssh"`
 - b. `export GIT_TRACE=1`
8. Push it into the main repository using "git push"

Process of porting redis to the zopen community:

Week 1:

- Worked on finding dependencies that redis used to work.
 - In order to do this, we needed to go to the redis Github and find the Install file, which normally contains these dependencies. In this case, we were redirected to the README.md file.
 - Once we saw all of the dependencies from Github, we put them into our buildenv file which will allow us to build this tool through z/OS.
 - In the file, it looks like this:
 - `export ZOPEN_STABLE_DEPS="(make openssl which pkgconfig check_python"`
 - Inside the parenthesis, we add the dependencies that are needed to compile this tool

- From there, we made sure that all of the dependencies were supported by z/OS which some were not so we had to exclude them from the list.
- Once that was set up, we wanted to build this tool to see if there were any errors that we might encounter that weren't the missing dependencies.
- Unfortunately, there were some errors in redis's source files that we needed to fix.
 - This would have to do with how these files compile in z/OS and how it would ignore flags placed to streamline the file-building process.

Week 3/11 - 3/14:

- I went to build the redis port and found these errors:
 - clang-14: error: no such file or directory: `'../deps/hdr_histogram/libhdrhistogram.a'`
 - clang-14: error: no such file or directory: `'../deps/fpconv/libfpconv.a'`
 - clang-14: error: no such file or directory: `'../deps/fast_float/libfast_float.a'`
- When looking for these files using the "find" command, we saw that it wasn't created.
 - This was due to the files not compiling in our buildenv.
- To fix this, we changed our file and added a new template that would allow us to change how MAKE works.
 - This is how it looks:
 - `export ZOPEN_MAKE="zopen_make"`
- From there we defined what the make should take in and reran our program.
 - Here are the specified dependencies:
 - `zopen_make() {`
 - `cd deps; make "$@" fast_float fpconv hdr_histogram hiredis jemalloc linenoise lua; cd -`
 - `make "$@"`
 - `}`
- This temporarily fixed the problem of the files not being created, but a new error was created in the fast_float file where one of the defined functions has an error.
- To troubleshoot this, we added two checks for the defined variable which would display an error depending on which defined variable failed first.

- Given this, we looked into every instance of the variable in our paths and found that it needed to run C++14. We then went into our buildenv so that when we built the program, the files would compile as C++14.
- This didn't fix our problem since it seemed that it was reverted to C++11 when compiling. We then found that in a file that redis was using to compile, it was hardcoded to be C++11 so we changed it to see if it would fix it and luckily it did.
- From there, we got a new error about the mutex not being included in one of the files. We looked at the file where the error was coming from and found that the Pthread initializer has a condition that checks if it is greater than 2 but since we set `-D_OPEN_THREADS` to 2, it wouldn't run. In order to fix this, we set it to 3 so that it could run.
 - This is how it looked:
 - `export CFLAGS="$CFLAGS $CPPFLAGS -mzos-target=zosv2r5 -D__XPLAT -D_OPEN_THREADS=3 -D__thread=-DMAP_ANON=0 -D_UNIX03`
- We tried to build redis once again and found that the files were still not being created causing an error. This was weird since it was making these files before. These files are made in the buildenv and we found that the "make "@"" caused these files to not be made.
 - Here is the code:
 - `zopen_make() {`
 - `cd deps; make "$@" fast_float fpconv hdr_histogram hiredis`
 - `linenoise lua; cd -`
 - `make "$@"`
 - `}`
- We will continue to work on this next week.

Week 3/18 - 3/21:

- At the beginning of the week, I worked on fixing an issue with the zopen manager.
 - The project coordinator had a meeting with us about the steps of fixing the issue, which will be shown here:
 - First we would need to update the cache to include the build dependency information (in

https://raw.githubusercontent.com/ZOSOpenTools/meta/main/docs/api/zopen_releases.json), then we can modify the zopen-info script to query for the build dependencies.

- Alternatively, the -r code can download the metadata.json for the given release and use it to grab the build dependencies. This would be my preference because if we add more information to the zopen_releases.json it will bloat it and may slow things down.
- I fixed a part of this issue and then passed on the work to my teammate so they could look into this issue as well.
- From there, I looked into another issue: a link from the zopen community website that led to a 404 webpage.
 - To fix this, I first looked into the text that the link was attached to which said:
 - “Setting up the CI/CD pipeline”
 - I then used the search bar in the zopen community website and typed in “CI/CD” to see all the webpages that had these words.
 - After a few webpages, I managed to find one that looked like it would fit the criteria that we set from the text from before.
 - To check if I was correct in my findings, I messaged my project coordinator, and luckily, it was correct.
 - I will add a PR to the issue next week.
- I worked on porting redis at the end of the week and tried to fix the issue from last week.
 - First, I tried adding the missing directories manually through the build env file by doing this:
 - `export ZOPEN_STABLE_DEPS="make openssl which pkgconfig check_python lua zoslib"`
 - `zopen_make() {`
 - `cd deps; make "$@" fast_float fpconv hdr_histogram hiredis linenoise lua zoslib; cd -`
 - `make "$@"`
 - `}`
 - This, however, did not change anything since the directories were still being deleted somehow.
 - After this didn't work, I came up with some other ways that I could try to fix the issue
 - I want to check if the libraries such as zoslib is properly linked
 - I want to check if the path is specified in the LDFLAGS

- I want to check if there is an error in the makefile that is causing these issues
 - I plan to continue working on this next week
- My project coordinator is going to be out of the office for the next week so I will have to make sure that I continue working on porting redis to z/OS.

Week 3/25 - 3/28:

- I continued to try to fix the build issue from last week
 - I looked into the Makefile to see if there was a reason why the make “@” was deleting the files after they were created.
 - I was specifically looking for any syntax errors that could have caused this and found that there was not an endif at the end of the file.
 - I then added it to the end of the Makefile. Then, when I ran it again, I was able to avoid that issue, but a new issue came up.
- The new issue that came up is this:
 - eval.c:249:24: error: use of undeclared identifier 'LUA_GLOBALSINDEX'
 - lua_pushvalue(lua, LUA_GLOBALSINDEX);
- This issue appears since redis is using an older version of lua, so “LUA_GLOBALSINDEX” isn’t compatible with the version z/OS is running
 - To fix this issue, I replaced “LUA_GLOBALSINDEX” with the equivalent variable in the newer version of lua
 - This is how it looks now:
 - lua_pushvalue(lua, LUA_RIDX_GLOBALS);
- I tried to build again, but a new error came up relating to this where another lua file needs to have this variable changed as well.
 - Here is one of the errors that popped up:
 - script_lua.c:1637:38: error: use of undeclared identifier 'LUA_GLOBALSINDEX'
 - lua_enablereadonlytable(lua, LUA_GLOBALSINDEX, 1);
 - The fix to this issue is the same as before, where we would replace the variable again.
- After fixing the issues on that file, I tried to build again only to be met with the same issue from another lua file.
 - Here is an example of one of the errors:

- function_lua.c:495:38: error: use of undeclared identifier 'LUA_GLOBALSINDEX'
 - lua_replace(lua_engine_ctx->lua, LUA_GLOBALSINDEX); /* set new global table as the new globals */
- Side Note: I used this command to find where the file was:
 - find /data/aquiles/redisport -name "function_lua.c"
- I replaced the variable again to try to fix the issue.
- After replacing the variable, I tried to build again and got a new issue:
 - IEW5034 The archive library libdl.a cannot be found.
 - clang-14: error: linker command failed with exit code 8 (use -v to see invocation)
 - This could be due to z/OS not supporting the library, so I need to find an alternative solution.

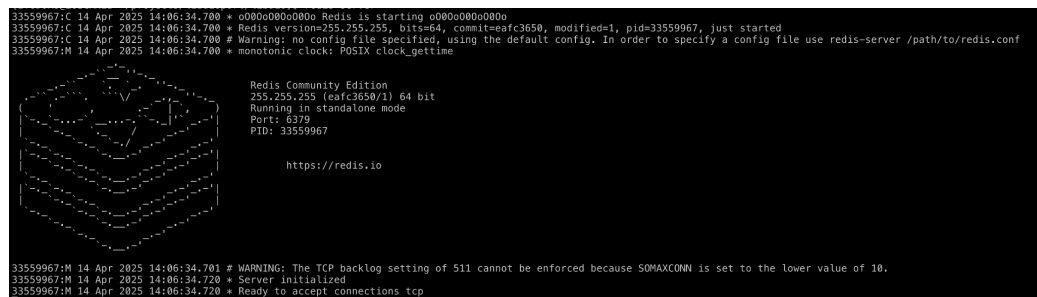
Week 4/1 - 4/4:

- Worked on first fully fixing the lua issue by first making sure that lua isn't a requirement for building redis then removing the dependency for it on the buildenv
 - zopen_make() {
 - cd deps; make "\$@" fast_float fpconv_hdr_histogram hiredis linenoise; cd -
 - make "\$@"
 - }
- We also looked into the main Makefile for lua and made a change where we removed any instance of -ldl since it is supported by a library that z/OS does not have.
- After, we tried to build it again and found that the errors that were found were from missing dependencies in lua.
 - This means that we need to change the lua version that is currently on redis and patch it with z/OS version, which should remove those errors.
- We found that the linker issue from before was due to the library dependencies not being created in the Makefile, so we added a line for that
- From there, we managed to begin building redis, and it went into the install phase, but there was an error that was due to the lua mismatch
 - Since we decided to change the lua version of z/OS to match with redis, we had to fix the changes I made with LUA_RIDX_GLOBALS.
 - We also went into the buildenv and added an extra dependency, "coreutils" which would allow redis to install when we are building it.

- Once we tried running it again, it managed to be a successful build, so we made a patch and committed it.
- From there, we began working on creating a release for it.
 - We did this by first going to this site: <https://cicd.zopen.community/>
 - This site allows us to add our project into the cicd pipeline
 - We then created a new job then used the redis github link so the process would go faster.
 - We filled out the criteria that were necessary from the zopen community, then we began to run it
- There was an issue with this where redis is referencing old os systems instead of the new ones so we will have to fix this next time.

Week 4/15 - 4/18:

- We managed to make a release for redis, so now users are able to use this open-source tool.
 - It can be used in z/OS in one of two ways:
 - `curl -o redis-unstable.20250409_011848.zos.pax.Z -L https://github.com/zopencommunity/redisport/releases/download/STABLE\_redisport\_3219/redis-unstable.20250409\_011848.zos.pax.Z && pax -rf redis-unstable.20250409_011848.zos.pax.Z && cd redis-unstable && . ./env`
 - `zopen install redis`
- Now that it can be used in z/OS I want to show some pictures of it working.



```

33559967:C 14 Apr 2025 14:06:34.700 * o00o00o0o00o Redis is starting o00o00o0o00o
33559967:C 14 Apr 2025 14:06:34.700 * Redis version=255.255.255, bits=64, commit=ea3c3650, modified=1, pid=33559967, just started
33559967:C 14 Apr 2025 14:06:34.700 * Warning: no config file specified, using the default config. In order to specify a config file use redis-server /path/to/redis.conf
33559967:M 14 Apr 2025 14:06:34.700 * monotonic clock: POSIX clock_gettime

Redis Community Edition
255.255.255 (ea3c3650/1) 64 bit
Running in standalone mode
Port: 6379
PID: 33559967

https://redis.io

33559967:M 14 Apr 2025 14:06:34.701 # WARNING: The TCP backlog setting of 511 cannot be enforced because SO_MAXCONN is set to the lower value of 10.
33559967:M 14 Apr 2025 14:06:34.720 * Server initialized
33559967:M 14 Apr 2025 14:06:34.720 * Ready to accept connections tcp
  
```

- - In this photo, we use the command `redis-server`, which will allow us to run a server on our terminal. This will allow us to access databases through redis.

```
[ITODOR0@Z0SCAN2B ~]$ redis-cli
127.0.0.1:6379> SET tutorialspoint redis
OK
127.0.0.1:6379> GET tutorialspoint
"redis"
127.0.0.1:6379> LPUSH tutorials redis
(integer) 1
127.0.0.1:6379> LPUSH tutorials mongodb
(integer) 2
127.0.0.1:6379> LRANGE tutorials 0 10
1) "mongodb"
2) "redis"
127.0.0.1:6379> █
```

- - These are commands run in a second terminal while the server is running.
 - Here is what each command does:
 - SET tutorialspoint redis: Creates or updates the key tutorialspoint with the value redis. It's useful for storing simple key-value pairs.
 - GET tutorialspoint: Retrieves the value associated with the key tutorialspoint. This confirms that the SET command worked.
 - LPUSH tutorials redis: Adds redis to the beginning of a list named tutorials.
 - LPUSH tutorials mongodb: Adds another item, mongodb, to the beginning of the list. The list now contains both items, in reverse order of insertion.
 - LRANGE tutorials 0 10: Fetches items from the tutorials list, starting at index 0 and ending at index 10. In this case, it retrieves the two items mongodb and redis.
- So, what does redis allow for in z/OS?
 - It allows for rapid data sharing across platforms, real-time analytics, and advanced key-value storage for flexible data structures.
- Hopefully, this open-source tool will be useful to many developers as time passes.

